Vinit Nagar

BACKEND & AI SYSTEMS ENGINEER

Jodhpur, India — +91 81122 48984 — vinitnagar56@gmail.com — github.com/vinitngr — linkedin.com/in/vinit-nagar-264434293

About

Final-year student at MBM University focused on backend engineering, AI systems, and inference optimization. I build distributed, production-grade applications, from multi-agent RAG pipelines to event-driven microservices. Working primarily in TypeScript, Go, and Python, I prioritize building fast, maintainable systems engineered for real-world production constraints.

Experience

Hooman Digital — *Sep 2025 - Present*

SDE Intern - AI & Backend (Remote)

- Engineered a production multi-agent RAG system with dynamic tool execution, replacing a basic chat interface and streamlining context retrieval for end users.
- Architected a scalable microservices backend with an API Gateway using gRPC for low-latency inter-service communication and RabbitMQ for asynchronous task distribution.
- Developed high-performance Go workers for compute-heavy ingestion and summarization pipelines, offloading blocking operations from the main application thread.
- Automated GPU orchestration using the Nosana SDK and MCP, enabling one-click AI model deployment through AI-generated server configurations.

Wisemango Inc. — *Jul 2025 - Sep 2025*

Full-Stack Developer (Remote)

- Implemented secure OAuth2 authentication flows for the Klype SaaS platform using NextAuth.
- Built concurrent data scraping pipelines in Node.js to automate lead generation, processing thousands of records per run.
- Optimized server-side rendering for high-traffic Next.js pages, improving Core Web Vitals and lowering TTFB.

Achievement

Winner - AWS Global Hackathon (4x Track)

Built Codrel, a distributed context analysis engine for AI-assisted coding. Selected across 4 distinct challenge tracks, achieving the highest number of track wins among all global participants.

Projects

Codrel - Distributed Context Analysis Engine — *AWS Hackathon Winner*

TypeScript, Go, Kafka, MCP, Vector DB, LLM Integration — github.com/vinitngr/codrel

- Designed a distributed event-streaming pipeline using Kafka to ingest and process high-velocity Git repository data in real time.
- Wrote concurrent Go workers for parallel code analysis, summarization, and vector ingestion, reducing pipeline processing latency.
- Integrated an MCP server to instantly surface processed context to LLM agents during live coding sessions, improving suggestion relevance.

RTCboard - Real-time Collaborative Whiteboard — *WebRTC, P2P*

WebSockets, WebRTC, Redis Pub/Sub, Node.js — github.com/vinitngr/rtcboard

- Built a peer-to-peer collaborative whiteboard achieving sub-100ms latency using WebSockets and WebRTC for real-time data transport.
- Implemented Redis pub/sub for ephemeral state synchronization across distributed backend nodes, supporting multiple concurrent sessions.

Nodebox - Browser-based Container Runtime — *Edge, Sandboxing*

WebContainers, Cloudflare Workers, Virtual File System — github.com/vinitngr/nodebox

- Engineered a browser-based Node.js runtime using WebContainers with a virtualized file system for secure, sandboxed in-browser code execution.
- Deployed on Cloudflare Workers for edge-optimized delivery, minimizing cold-start latency globally.

Technical Skills

Languages	TypeScript, Go, Python, SQL
Backend & APIs	System Design, Node.js, REST API, Next.js, Express.js
Infrastructure	Docker, AWS (EC2, S3), Cloudflare Workers, Linux
Messaging & Data	RabbitMQ, Redis, PostgreSQL, MongoDB
AI Engineering	RAG Pipelines, Agentic Workflows, MCP, Vector Databases, n8n
Tools	Git, GitHub, WebRTC, Postman

Education

B.E. Electronics & Communication Engineering *2023 - 2027*

MBM University, Jodhpur

Relevant coursework: Operating Systems, Computer Networks, DBMS, Data Structures & Algorithms